Double Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Double integrals over general regions**

Evaluate $\int_0^1 \int_0^x (x+y)\,dy\,dx$, integrating over the triangular region $0 \le y \le x \le 1$.

Inner: $\int_0^x (x+y)\,dy = x^2 + \frac{x^2}{2} = \frac{3x^2}{2}$. Outer: $\int_0^1 \frac{3x^2}{2}\,dx = \left[\frac{x^3}{2}\right]_0^1 = \frac{1}{2}$.

2 **[Foundational]** • **Target: Double integrals over general regions**

Use a double integral to find the volume under $f(x, y) = 4 - x - y$ over the rectangle $0 \le x \le 1$, $0 \le y \le 1$.

$\int_0^1 \int_0^1 (4 - x - y)\,dy\,dx$: inner $= 4 - x - \frac{1}{2}$. Outer: $\int_0^1 \left(\frac{7}{2} - x\right)dx = \frac{7}{2} - \frac{1}{2} = 3$.

3 **[Foundational]** • **Target: Reversing the order of integration**

Evaluate $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$ by first reversing the order of integration (the region is $0 \le y \le x \le 1$).

Reversing: $\int_0^1 \int_0^x e^{x^2}\,dy\,dx = \int_0^1 x e^{x^2}\,dx = \left[\frac{1}{2}e^{x^2}\right]_0^1 = \frac{e-1}{2}$.

4 **[Foundational]** • **Target: Double integrals in polar coordinates**

Evaluate $\iint_D (x^2 + y^2)\,dA$ where D is the disk $x^2 + y^2 \le 4$, by converting to polar coordinates ($dA = r\,dr\,d\theta$).

$\int_0^{2\pi} \int_0^2 r^2 \cdot r\,dr\,d\theta = \int_0^{2\pi} \left[\frac{r^4}{4}\right]_0^2 d\theta = \int_0^{2\pi} 4\,d\theta = 8\pi$.

5 **[Foundational]** • **Target: Double integrals in polar coordinates**

Find the area of the region enclosed by $r = 2\cos\theta$ using a double integral in polar coordinates.

$A = \int_{-\pi/2}^{\pi/2} \int_0^{2\cos\theta} r\,dr\,d\theta = \int_{-\pi/2}^{\pi/2} 2\cos^2\theta\,d\theta = \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta)\,d\theta = \pi$.

6 **[Foundational]** • **Target: Average value over a region**

Find the average value of $f(x, y) = xy$ over the rectangle $0 \le x \le 2$, $0 \le y \le 3$ (average value $= \frac{1}{A}\iint_D f\,dA$, where A is the region's area).

$\iint_D xy\,dA = \int_0^2 \int_0^3 xy\,dy\,dx = \int_0^2 x \cdot \frac{9}{2}\,dx = \frac{9}{2} \cdot 2 = 9$. Area $= 6$. Average $= \frac{9}{6} = \frac{3}{2}$.

7 **[Foundational]** • **Target: Mass and center of a region**

A lamina occupies the rectangle $0 \le x \le 2$, $0 \le y \le 1$ with density $\rho(x, y) = x + y$. Find its total mass, $M = \iint_D \rho\,dA$.

$M = \int_0^2 \int_0^1 (x + y)\,dy\,dx = \int_0^2 \left(x + \frac{1}{2}\right)dx = \left[\frac{x^2}{2} + \frac{x}{2}\right]_0^2 = 2 + 1 = 3$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Mass and center of a region**

For a lamina occupying $0 \le x \le 1$, $0 \le y \le 1$ with constant density $\rho = 1$, find the x-coordinate of the center of mass, $\bar{x} = \frac{1}{M}\iint_D x\,dA$ (where M is the total mass).

$M = 1 \cdot 1 = 1$. $\iint_D x\,dA = \int_0^1 \int_0^1 x\,dy\,dx = \int_0^1 x\,dx = \frac{1}{2}$. $\bar{x} = \frac{1/2}{1} = \frac{1}{2}$ (as expected, the center of a uniform square).

9 **[Intermediate]** • **Target: Triple integral setup (a first look)**

Set up (do not evaluate) the triple integral for the volume of the solid bounded by $0 \le x \le 1$, $0 \le y \le 2$, $0 \le z \le 3$.

$V = \int_0^1 \int_0^2 \int_0^3 dz\,dy\,dx$ (which evaluates to $1 \times 2 \times 3 = 6$, the box's volume).

10 **[Intermediate]** • **Target: Double integrals with a variable upper limit**

Evaluate $\int_0^1 \int_0^{y^2} 3x\,dx\,dy$, integrating over the region $0 \le x \le y^2$, $0 \le y \le 1$.

Inner: $\int_0^{y^2} 3x\,dx = \frac{3}{2}y^4$. Outer: $\int_0^1 \frac{3}{2}y^4\,dy = \left[\frac{3}{10}y^5\right]_0^1 = \frac{3}{10}$.

11 **[Intermediate]** • **Target: Double integrals with a variable upper limit**

Set up and evaluate the double integral for the volume under $f(x, y) = xy^2$ over the triangular region bounded by $y = x$, $y = 0$, and $x = 2$.

$V = \int_0^2 \int_0^x xy^2\,dy\,dx = \int_0^2 x \cdot \frac{x^3}{3}\,dx = \int_0^2 \frac{x^4}{3}\,dx = \left[\frac{x^5}{15}\right]_0^2 = \frac{32}{15}$.

12 **[Intermediate]** • **Target: An applied double-integral problem**

A metal plate occupies the region $0 \le x \le 3$, $0 \le y \le 2$, with temperature distribution $T(x, y) = 50 - x^2 - y^2$ (degrees Celsius). Find the average temperature over the plate.

$\iint_D T\,dA = \int_0^3 \int_0^2 (50 - x^2 - y^2)\,dy\,dx = \int_0^3 \left(100 - 2x^2 - \frac{8}{3}\right)dx = \left[\frac{292}{3}x - \frac{2x^3}{3}\right]_0^3 = 292 - 18 = 274$. Area $= 6$. Average $= \frac{274}{6} \approx 45.67$°C.

13 **[Intermediate]** • **Target: Volume between two surfaces**

Find the volume of the solid between $z = 8 - x^2 - y^2$ and $z = 0$ over the region where $z \ge 0$ requires $x^2 + y^2 \le 8$, using polar coordinates.

$V = \int_0^{2\pi} \int_0^{2\sqrt{2}} (8 - r^2) r\,dr\,d\theta = \int_0^{2\pi} \left[4r^2 - \frac{r^4}{4}\right]_0^{2\sqrt{2}} d\theta = \int_0^{2\pi} (32 - 16)\,d\theta = \int_0^{2\pi} 16\,d\theta = 32\pi$.

14 **[Intermediate]** • **Target: Double integrals with exponential integrands**

Evaluate $\int_0^1 \int_0^1 xe^{xy}\,dy\,dx$ by integrating with respect to y first.

Inner: $\int_0^1 xe^{xy}\,dy = [e^{xy}]_0^1 = e^x - 1$ (for $x \neq 0$). Outer: $\int_0^1 (e^x - 1)\,dx = [e^x - x]_0^1 = (e - 1) - 1 = e - 2$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: A capstone applied double-integral problem** • **Cross-Lesson: 8.03 Arc Length and Improper Integrals**

An architect is designing a swimming pool whose floor depth varies according to $d(x, y) = 1 + 0.2x + 0.1y$ (meters) over a rectangular footprint $0 \le x \le 10$, $0 \le y \le 5$ (meters). Set up and evaluate the double integral for the total volume of water needed to fill the pool, then find the average depth by dividing by the pool's surface area.

$V = \int_0^{10}\int_0^5 (1 + 0.2x + 0.1y)\,dy\,dx$. Inner: $\int_0^5 (1 + 0.2x + 0.1y)\,dy = 5 + x + 1.25 = 6.25 + x$. Outer: $\int_0^{10} (6.25 + x)\,dx = \left[6.25x + \frac{x^2}{2}\right]_0^{10} = 62.5 + 50 = 112.5$ m^3. Surface area $= 10 \times 5 = 50$ m^2. Average depth $= \frac{112.5}{50} = 2.25$ m.

16 **[Challenging]** • **Target: A capstone applied double-integral problem** • **Cross-Lesson: 9.01 Separable Differential Equations**

For the double integral $\iint_D (2x + y)\,dA$ where D is the region bounded by $y = x^2$ and $y = x$, complete each step:

a Find the intersection points of the two curves to determine the bounds. $x^2 = x$ gives $x = 0, 1$. On $[0, 1]$, $x \ge x^2$, so $x^2 \le y \le x$.

b Set up the iterated integral with x as the outer variable. $\iint_D (2x + y)\,dA = \int_0^1\int_{x^2}^{x} (2x + y)\,dy\,dx$.

c Evaluate the integral. Inner: $\left[2xy + \frac{y^2}{2}\right]_{x^2}^{x} = \left(2x^2 + \frac{x^2}{2}\right) - \left(2x^3 + \frac{x^4}{2}\right) = \frac{5x^2}{2} - 2x^3 - \frac{x^4}{2}$. Outer: $\int_0^1 \left(\frac{5x^2}{2} - 2x^3 - \frac{x^4}{2}\right)dx = \frac{5}{6} - \frac{1}{2} - \frac{1}{10} = \frac{25 - 15 - 3}{30} = \frac{7}{30}$.

(a) $x^2 = x$ gives $x = 0, 1$. On $[0, 1]$, $x \ge x^2$, so $x^2 \le y \le x$.; (b) $\iint_D (2x + y)\,dA = \int_0^1\int_{x^2}^{x} (2x + y)\,dy\,dx$.; (c) Inner: $\left[2xy + \frac{y^2}{2}\right]_{x^2}^{x} = \left(2x^2 + \frac{x^2}{2}\right) - \left(2x^3 + \frac{x^4}{2}\right) = \frac{5x^2}{2} - 2x^3 - \frac{x^4}{2}$. Outer: $\int_0^1 \left(\frac{5x^2}{2} - 2x^3 - \frac{x^4}{2}\right)dx = \frac{5}{6} - \frac{1}{2} - \frac{1}{10} = \frac{25 - 15 - 3}{30} = \frac{7}{30}$.

17 **[Challenging]** • **Target: Average value over a region** • **Cross-Lesson: 1.01 Precalculus Review for Calculus**

Level 2 · Medium Evaluate the iterated integral by reversing the order of integration: $\int_0^{\pi}\int_y^{\pi} \frac{\sin x}{x}\,dx\,dy$

The domain is $0 \le y \le \pi$ and $y \le x \le \pi$. This is the triangular region bounded by $y = 0$, $x = \pi$, and $y = x$. Reversing order to Type I: x ranges from 0 to π, while y ranges from 0 to x: $\int_0^{\pi}\int_0^{x} \frac{\sin x}{x}\,dy\,dx = \int_0^{\pi}\left[y\frac{\sin x}{x}\right]_{y=0}^{y=x} dx = \int_0^{\pi} x\frac{\sin x}{x}\,dx = \int_0^{\pi} \sin x\,dx = [-\cos x]_0^{\pi} = -\cos(\pi) - (-\cos 0) = -(-1) - (-1) = 1 + 1 = 2$

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18 [Challenging] • Target: Mass and center of a region • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Level 3 · Advanced Evaluate the double integral $\iint_D e^{x^2+y^2}\,dA$ where D is the semi-annular region $\{(x,y): 1 \le x^2+y^2 \le 4, y \ge 0\}$.

Convert the semi-annulus to polar coordinates: Radius bounds: $1 \le r^2 \le 4 \Rightarrow 1 \le r \le 2$. Angle bounds (since $y \ge 0$, upper half-plane): $0 \le \theta \le \pi$. Area element: $dA = r\,dr\,d\theta$. Set up the integral: $\iint_D e^{x^2+y^2}\,dA = \int_0^{\pi}\int_1^2 e^{r^2}\,r\,dr\,d\theta = \left(\int_0^{\pi} d\theta\right)\left(\int_1^2 re^{r^2}\,dr\right)$ Substitute $u = r^2 \Rightarrow du = 2r\,dr$.

19 [Challenging] • Target: Triple integral setup (a first look) • Cross-Lesson: 11.01 Sequences and Series

Level 2 · Medium Use the method of cylindrical shells to find the volume generated by revolving the region bounded by $y = 3x - x^2$ and $y = 0$ about the y-axis.

Find intersections with $y = 0$: $3x - x^2 = 0 \Rightarrow x(3-x) = 0$, so $a = 0$ and $b = 3$. Revolving about y-axis: shell radius $r(x) = x$, shell height $h(x) = 3x - x^2$. $V = 2\pi\int_0^3 x(3x-x^2)\,dx = 2\pi\int_0^3 (3x^2 - x^3)\,dx = 2\pi\left[x^3 - \frac{x^4}{4}\right]_0^3$ $= 2\pi\left(27 - \frac{81}{4}\right) = 2\pi\left(\frac{27}{4}\right) = \frac{27\pi}{2}$

20 [Challenging] • Target: Double integrals with a variable upper limit • Cross-Lesson: 12.01 Convergence Tests

Level 3 · AdvancedEvaluate the definite integral analytically: $\int_{-3}^{3}\left(x^5\sqrt{9-x^2} + \sqrt{9-x^2}\right)dx$.

Split the integral using linearity: $\int_{-3}^{3} x^5\sqrt{9-x^2}\,dx + \int_{-3}^{3}\sqrt{9-x^2}\,dx$. Part 1 · Symmetry Analysis: Let $g(x) = x^5\sqrt{9-x^2}$. Then $g(-x) = (-x)^5\sqrt{9-(-x)^2} = -x^5\sqrt{9-x^2} = -g(x)$. Because g is an odd function integrated over symmetric limits $[-3, 3]$: $\int_{-3}^{3} x^5\sqrt{9-x^2}\,dx = 0$. Part 2 · Geometric Area: The integrand $y = \sqrt{9-x^2} \ge 0$ represents the upper half of the circle $x^2 + y^2 = 9$ with radius $r = 3$. The integral from.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Iterated integrals over rectangles

Consider the solid E bounded by the surfaces $z = x^2 + y^2$ and $z = 8 - x^2 - y^2$. Calculate the volume of this solid using a triple integral by first transforming to cylindrical coordinates. Express your answer in terms of π.

a. $\frac{16\pi}{3}$

b. $\frac{32\pi}{3}$ ✓

c. 16π

d. $\frac{64\pi}{3}$

The solid E is bounded below by the paraboloid $z = x^2 + y^2$ and above by the paraboloid $z = 8 - x^2 - y^2$. To find the intersection of these two surfaces, we set them equal: $x^2 + y^2 = 8 - x^2 - y^2$, which simplifies to $2(x^2 + y^2) = 8$, or $x^2 + y^2 = 4$. This is a circle of radius 2 in the xy-plane. In cylindrical coordinates, $x = r\cos(\theta)$, $y = r\sin(\theta)$, and $z = z$. The equations of the surfaces become $z = r^2$ and $z = 8 - r^2$. The intersection occurs at $r^2 = 8 - r^2$, so.

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22 [Foundational] • Target: Double integrals over general regions

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?

a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$ ✓

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

The region R is bounded above by $y = 4$ and below by $y = x^2$. To find the limits for x, we find the intersection points of the two curves: $x^2 = 4 \Rightarrow x = \pm 2$. Thus, the region is defined for x from -2 to 2. When integrating with respect to y first ($dydx$), the inner integral goes from the lower curve to the upper curve, so from $y = x^2$ to $y = 4$. The outer integral is over the range of x, which is -2 to 2. Therefore, the double integral representing the area is.

23 [Foundational] • Target: Reversing the order of integration

Iterated integral over tetrahedron $x + y + z \le 1, x, y, z \ge 0$: volume = $1/6$. The integral $\iiint x\, dV =$

a. $1/6$

b. $1/24$ ✓

c. $1/12$

d. $1/8$

By symmetry $\iiint x = \iiint y = \iiint z$; and $\iiint (x + y + z) = 3\iiint x$; by integration $\iiint (x + y + z)dV = 1/8$... actually $\iiint x\, dV = 1/24$ (standard result).

24 [Foundational] • Target: Double integrals in polar coordinates

The region between $y = x$ and $y = x^3$ in the first quadrant is revolved about the y-axis. Find the volume.

a. $\frac{4\pi}{21}$

b. $\frac{4\pi}{15}$ ✓

c. $\frac{2\pi}{15}$

d. $\frac{4}{15}$

The curves meet at $x = 0$ and $x = 1$. Shells: $V = 2\pi\int_{0}^{1} x(-x^3 + x)\, dx = \frac{4\pi}{15}$.

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25 [Foundational] • Target: Average value over a region

The double integral over region D of $f = 1$ gives:

a. area of D ✓

b. perimeter of D

c. volume

d. 0

Applying Multiple Integrals principles, $\iint_D 1\,dA = \text{Area}(D)$. Therefore, the verified correct option is area of D.

26 [Foundational] • Target: Mass and center of a region

Evaluate $\int_0^2 \int_y^2 e^{x^2}\,dx\,dy$ by reversing the order of integration.

a. $-\frac{1}{2} + \frac{e^4}{2}$ ✓

b. $-1 + e^4$

c. $\frac{1}{2} + \frac{e^4}{2}$

d. $-\frac{1}{4} + \frac{e^4}{4}$

The inner integral has no elementary antiderivative as written. Reversing the order gives $\int_0^2 \int_0^x e^{x^2}\,dy\,dx$, which evaluates to $-\frac{1}{2} + \frac{e^4}{2}$.

27 [Foundational] • Target: Triple integral setup (a first look)

The volume of the region $1 \le x^2 + y^2 + z^2 \le 4$ (shell between spheres of radii 1 and 2):

a. $4\pi/3$

b. 4π

c. 8π

d. $4\pi(8-1)/3 = 28\pi/3$ ✓

Applying Multiple Integrals principles, $V = (4\pi/3)(2^3 - 1^3) = 28\pi/3$. Therefore, the verified correct option is $4\pi(8-1)/3 = 28\pi/3$.

28 [Foundational] • Target: Double integrals with a variable upper limit

Use spherical coordinates to find the volume of the first-octant part of the ball $x^2 + y^2 + z^2 \le 16$.

a. $\frac{256\pi}{3}$

b. $\frac{64\pi}{3}$

c. $\frac{32\pi}{3}$ ✓

d. $\frac{16\pi}{3}$

$dV = \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$; integrating over the region gives $\frac{1}{8} \cdot \frac{4}{3}\pi(4)^3 = \frac{32\pi}{3}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: An applied double-integral problem

Volume under $z = 1$ over region R of area A is:

a. A^2

b. A ✓

c. $A/2$

d. πA

Applying Applications of Integration principles, $V = \iint_R 1\, dA = A$. Therefore, the verified correct option is A.

30 [Intermediate] • Target: Volume between two surfaces

Use polar coordinates to evaluate $\iint_R xy\, dA$, where R is the part of the disk $x^2 + y^2 \le 16$ in the first quadrant.

a. 32 ✓

b. 32π

c. 64

d. 16

With $x = r\cos\theta,\ y = r\sin\theta$ and $dA = r\, dr\, d\theta$, the integral becomes elementary; its value is 32.

31 [Intermediate] • Target: Double integrals with exponential integrands

Volume between $z = x^2 + y^2$ (paraboloid) and $z = 1$ over disk $r \le 1$:

a. $3\pi/4$

b. π

c. $\pi/4$

d. $\pi/2$ ✓

$V = \int_0^{2\pi}\int_0^1 (1 - r^2) r\, dr\, d\theta = 2\pi(1/2 - 1/4) = \pi/2.$

32 [Intermediate] • Target: A capstone applied double-integral problem

Volume between $z = 1$ and $z = x^2 + y^2$ over unit disk:

a. $\pi/4$

b. $\pi/2$ ✓

c. π

d. $3\pi/4$

$V = \iint_{x^2+y^2\le 1} (1 - (x^2 + y^2))dA = \pi \cdot 1 - \pi/2 = \pi/2.$

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33 [Intermediate] • Target: Iterated integrals over rectangles

Use a double integral to find the area of the region under $y = x^2$ for $0 \le x \le 1$.

a. $\frac{2}{3}$

b. $\frac{1}{3}$ ✓

c. $\frac{1}{6}$

d. $\frac{1}{2}$

$\int_0^1 \int_0^{x^2} 1 \, dy \, dx = \frac{1}{3}$.

34 [Intermediate] • Target: Double integrals over general regions

Volume between planes $z = 0$ and $z = 3x + 2y$ over $D = [0, 1] \times [0, 1]$:

a. 5/2 ✓

b. 3

c. 1

d. 3/2 + 1 = 5/2

$\int_0^1 \int_0^1 (3x + 2y) dy \, dx = \int_0^1 (3x + 1) dx = 3/2 + 1 = 5/2$.

35 [Intermediate] • Target: Reversing the order of integration

Volume integral: using spherical coordinates, $dV =$

a. $dr \, d\theta \, d\phi$

b. $r^2 \sin\phi \, dr \, d\theta \, d\phi$ ✓

c. $r \sin\theta \, dr \, d\theta \, d\phi$

d. $r^2 \, dr \, d\theta \, d\phi$

Spherical: $dV = r^2 \sin\phi \, dr \, d\theta \, d\phi$.

36 [Intermediate] • Target: Double integrals in polar coordinates

Volume under $z = f(x, y)$ over region D equals:

a. $\iint_D f^2 \, dA$

b. $\oint_D f \, ds$

c. $\iint_D ablaf \, dA$

d. $\iint_D f \, dA$ ✓

Volume = double integral of height.

37 [Intermediate] • Target: Average value over a region

Volume between paraboloid $z = x^2 + y^2$ and $z = 1$ over unit disk:

a. $3\pi/4$

b. π

c. $\pi/4$

d. $\pi/2$ ✓

$V = 2\pi \int_0^1 (1 - r^2) r \, dr = \pi/2$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Mass and center of a region** • **Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves**

For iterated integral $\int_a^b \left(\int_{g_1(x)}^{g_2(x)} f(x,y)dy \right) dx$: the inner integral is done first, then the outer. True or false?

a. False — outer first

b. **True ✓**

c. Only for separable functions

d. Depends on Fubini

Inner integral is evaluated first.

39 **[Challenging]** • **Target: Triple integral setup (a first look)** • **Cross-Lesson: 11.01 Sequences and Series**

Consider a region in the first quadrant bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis. A rectangle is inscribed within this region such that one vertex is at the origin, two sides lie along the x and y axes, and the fourth vertex lies on the curve $y = 9 - x^2$. If the area of this inscribed rectangle is maximized, determine the value of the definite integral that represents the volume of a solid formed by revolving the region bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis about the y-axis.

a. $\frac{81\pi}{2}$ ✓

b. 81π

c. 162π

d. $\frac{243\pi}{2}$

Let the vertex of the inscribed rectangle on the curve be (x, y). Since the vertex is on $y = 9 - x^2$, and the rectangle is in the first quadrant, $x \geq 0$ and $y \geq 0$. The dimensions of the rectangle are x and y. The area of the rectangle is $A = xy$. Substituting $y = 9 - x^2$, we get $A(x) = x(9 - x^2) = 9x - x^3$ for $0 \leq x \leq 3$. To maximize the area, we find the derivative of $A(x)$ with respect to x and set it to zero: $A'(x) = 9 - 3x^2$ Setting $A'(x) = 0$ gives $9 - 3x^2 = 0$, so.

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40 [Challenging] • Target: Double integrals with a variable upper limit • Cross-Lesson: 12.01 Convergence Tests

Consider a region R in the first quadrant bounded by the curve $y = e^{-x^2}$, the x-axis, and the y-axis. Let S be the solid generated when R is revolved about the y-axis. The volume of S can be calculated using the method of cylindrical shells or the disk/washer method. However, direct application of standard integration techniques to find the volume of S leads to an integral that cannot be evaluated in terms of elementary functions. A related problem involves finding the surface area of a solid of revolution. Let C be the curve defined by $y = \sqrt{1 - x^2}$ for $0 \le x \le 1$. This curve represents a quarter-circle. If this curve C is revolved about the x-axis, it generates a hemisphere. The surface area of this hemisphere (excluding the base) is given by the integral $A = \int_0^1 2\pi y \sqrt{1 + (y')^2}\, dx$. Determine the exact surface area of the generated hemisphere.

a. $\frac{\pi}{2}$

b. π ✓

c. 2π

d. 4π

The curve is given by $y = \sqrt{1 - x^2}$ for $0 \le x \le 1$. We first find the derivative of y with respect to x: $y' = \frac{d}{dx}(\sqrt{1 - x^2}) = \frac{1}{2\sqrt{1 - x^2}} \cdot (-2x) = \frac{-x}{\sqrt{1 - x^2}}$. Next, we compute $1 + (y')^2$: $1 + (y')^2 = 1 + \left(\frac{-x}{\sqrt{1 - x^2}}\right)^2 = 1 + \frac{x^2}{1 - x^2} = \frac{1 - x^2 + x^2}{1 - x^2} = \frac{1}{1 - x^2}$. Then, $\sqrt{1 + (y')^2} = \sqrt{\frac{1}{1 - x^2}} = \frac{1}{\sqrt{1 - x^2}}$ for $0 \le x < 1$. The surface area integral is.

41 [Challenging] • Target: An applied double-integral problem • Cross-Lesson: 12.02 Power Series and Taylor Series

The figure shows the graph of a continuous function $f(x)$ on the interval $[a, d]$. The shaded region represents the definite integral $\int_a^c f(x)\, dx$. If the area of the shaded region is 12 square units and the area of the unshaded region between $x = c$ and $x = d$ is 5 square units, what is the value of $\int_a^d f(x)\, dx$?

a. **17** ✓

b. 7

c. 2.4

d. -7

The Fundamental Theorem of Calculus relates the definite integral of a function to its antiderivative. Geometrically, the definite integral $\int_a^b f(x)\, dx$ represents the net signed area between the curve $f(x)$ and the x-axis from $x = a$ to $x = b$. Areas above the x-axis are positive, and areas below the x-axis are negative. The problem states that the area of the shaded region, representing $\int_a^c f(x)\, dx$, is 12 square units. This implies that this region contributes positively to the net signe.

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42 [Challenging] • Target: Volume between two surfaces • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Volume of region bounded by $x^2/4 + y^2/9 = 1$ rotated about the x-axis:

a. 24π ✓

b. 12π

c. 48π

d. 36π

Rotating $\frac{x^2}{4} + \frac{y^2}{9} = 1$ about the x-axis gives $V = \pi\int_{-2}^{2} 9\left(1 - \frac{x^2}{4}\right) dx = 9\pi\left[x - \frac{x^3}{12}\right]_{-2}^{2} = 24\pi$.

43 [Challenging] • Target: Double integrals with exponential integrands • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Region $x \geq 0$, $y \geq 0$, $x + y \leq 1$: volume when rotated about y-axis by shell method:

a. $\frac{\pi}{3}$ ✓

b. $\frac{\pi}{4}$

c. $\frac{\pi}{2}$

d. $\frac{\pi}{6}$

By cylindrical shells about the y-axis, $V = 2\pi\int_{0}^{1} x(1 - x)\,dx = 2\pi\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_{0}^{1} = 2\pi\left(\frac{1}{6}\right) = \frac{\pi}{3}$.

44 [Challenging] • Target: A capstone applied double-integral problem • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Pappus's theorem: volume of solid of revolution = $2\pi\bar{x} \cdot A$ where $\bar{x}$ is the centroid of the region and A is the:

a. arc length

b. surface area

c. **area of the region** ✓

d. cross-sectional area

Applying Applications of Integration principles, Pappus's centroid theorem. Therefore, the verified correct option is area of the region.

45 [Challenging] • Target: Iterated integrals over rectangles • Cross-Lesson: 2.01 Evaluating Limits

Volume of region $y \geq x^2$, $y \leq 4$ rotated about $y = 4$ using washers:

a. $256\pi/15$

b. $128\pi/15$

c. $512\pi/15$ ✓

d. $64\pi/15$

Applying Applications of Integration principles, $V = \pi\int_{-2}^{2}((4 - x^2)^2)dx$. Therefore, the verified correct option is $512\pi/15$.

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